



ER SCHEMA DDL SCRIPT

Code :

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-- 1. VENDOR  
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CREATE TABLE Vendor (  
  vendor_id          SERIAL PRIMARY KEY,  
  company_name       VARCHAR(150)    NOT NULL,  
  vendor_type        VARCHAR(20)     NOT NULL  
  
  CHECK (vendor_type IN ('hotel', 'transport', 'package')),  
  gst_number         VARCHAR(20)     UNIQUE NOT NULL,  
  registered_address TEXT            NOT NULL,  
  website            VARCHAR(255),  
  contact_name       VARCHAR(100)    NOT NULL,  
  contact_phone      VARCHAR(15),  
  contact_email      VARCHAR(100),
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is_active          BOOLEAN          DEFAULT TRUE,
created_at         TIMESTAMP        DEFAULT CURRENT_TIMESTAMP
);

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-- 2. USER
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CREATE TABLE "User" (
user_id            SERIAL PRIMARY KEY,
full_name          VARCHAR(150)    NOT NULL,
email              VARCHAR(100)    UNIQUE NOT NULL,
phone              VARCHAR(15)      UNIQUE NOT NULL,
dob                DATE             NOT NULL,
home_city          VARCHAR(100),
account_status     VARCHAR(20)      DEFAULT 'active'
CHECK (account_status IN ('active', 'suspended', 'deleted'))
);

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-- 3. TRANSPORT LAYER
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CREATE TABLE Transport_Route (
route_id SERIAL PRIMARY KEY,
transport_mode VARCHAR(10) NOT NULL
CHECK (transport_mode IN ('flight', 'train', 'bus')),
origin_city VARCHAR(100) NOT NULL,
destination_city VARCHAR(100) NOT NULL,
vehicle_number VARCHAR(50),
vendor_id INT NOT NULL REFERENCES Vendor(vendor_id)
);

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CREATE TABLE Route_Schedule (
  schedule_id SERIAL PRIMARY KEY,
  route_id INT NOT NULL REFERENCES Transport_Route(route_id)
  ON DELETE CASCADE,
  departure_time TIME NOT NULL,
  arrival_time TIME NOT NULL,
  days_of_week VARCHAR(20), -- e.g. 'Mon,Tue,Wed'
  is_active BOOLEAN DEFAULT TRUE
);

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CREATE TABLE Schedule_Class (
  class_id SERIAL PRIMARY KEY,
  class_name VARCHAR(50) NOT NULL,
  total_seats INT NOT NULL,
  base_fare DECIMAL(10,2) NOT NULL
);

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CREATE TABLE Route_Schedule_Class (
  schedule_id INT REFERENCES Route_Schedule(schedule_id) ON
  DELETE CASCADE,
  class_id INT REFERENCES Schedule_Class(class_id) ON DELET
  E CASCADE,
  PRIMARY KEY (schedule_id, class_id)
);

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CREATE TABLE Schedule_Departure (
  departure_id SERIAL PRIMARY KEY,

  schedule_id INT NOT NULL,
  class_id INT NOT NULL,

  departure_date DATE NOT NULL,
  available_seats INT NOT NULL,
  dynamic_fare DECIMAL(10,2), -- overrides base_fare on pea
  k dates

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        status VARCHAR(20) DEFAULT 'scheduled',

        -- fix : composite foreign key
        FOREIGN KEY (schedule_id, class_id)
            REFERENCES Route_Schedule_Class(schedule_id, class_id)
        ON DELETE CASCADE,

        CHECK (status IN ('scheduled', 'cancelled', 'completed'))
    );

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-- 4. HOTELS
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CREATE TABLE Hotel (
    hotel_id SERIAL PRIMARY KEY,
    hotel_name VARCHAR(150) NOT NULL,
    city VARCHAR(100) NOT NULL,
    full_address TEXT NOT NULL,
    star_rating DECIMAL(2,1) CHECK (star_rating BETWEEN 1.0 AND 5.0),
    checkin_time TIME,
    checkout_time TIME,
    vendor_id INT NOT NULL REFERENCES Vendor(vendor_id)
);

CREATE TABLE Hotel_Amenity (
    hotel_id INT REFERENCES Hotel(hotel_id) ON DELETE CASCADE,
    amenity VARCHAR(100) NOT NULL,
    PRIMARY KEY (hotel_id, amenity)
);

CREATE TABLE Room_Category (

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room_category_id SERIAL PRIMARY KEY,
hotel_id INT NOT NULL REFERENCES Hotel(hotel_id)
ON DELETE CASCADE,
category_name VARCHAR(100) NOT NULL, -- e.g. Deluxe, Suite, S
tandard
max_guests INT NOT NULL,
price_per_night DECIMAL(10,2) NOT NULL,
total_rooms INT NOT NULL
);

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CREATE TABLE Room_Availability (
room_category_id INT REFERENCES Room_Category(room_category_i
d)
ON DELETE CASCADE,
available_date DATE NOT NULL,
available_rooms INT NOT NULL,
PRIMARY KEY (room_category_id, available_date)
);

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-- 5. HOLIDAY PACKAGES
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CREATE TABLE Holiday_Package (
package_id SERIAL PRIMARY KEY,
package_name VARCHAR(200) NOT NULL,
theme VARCHAR(100), -- e.g. Adventure, Honeymoon
duration_days INT NOT NULL,
duration_nights INT NOT NULL,
base_price_per_person DECIMAL(10,2) NOT NULL,
is_active BOOLEAN DEFAULT TRUE,
vendor_id INT NOT NULL REFERENCES Vendor(vendor_id)
);

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CREATE TABLE Package_Hotel (
package_id INT REFERENCES Holiday_Package(package_id) ON DELETE CASCADE,
hotel_id INT REFERENCES Hotel(hotel_id) ON DELETE CASCADE,
PRIMARY KEY (package_id, hotel_id)

);

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CREATE TABLE Package_Transport (
package_id INT REFERENCES Holiday_Package(package_id) ON DELETE CASCADE,
departure_id INT REFERENCES Schedule_Departure(departure_id) ON DELETE CASCADE,
PRIMARY KEY (package_id, departure_id)
);

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CREATE TABLE Package_Departure (
package_departure_id SERIAL PRIMARY KEY,
package_id INT NOT NULL
REFERENCES Holiday_Package(package_id) ON DELETE CASCADE,
departure_date DATE NOT NULL,
total_seats INT NOT NULL,
available_seats INT NOT NULL,
price_per_person DECIMAL(10,2) NOT NULL,
status VARCHAR(20) DEFAULT 'open'
CHECK (status IN ('open', 'full', 'cancelled', 'completed'))
);

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-- 6. PROMO CODES
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CREATE TABLE Promo_Code (
promo_id SERIAL PRIMARY KEY,

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code                VARCHAR(50)        UNIQUE NOT NULL,
discount_type       VARCHAR(20)        NOT NULL
CHECK (discount_type IN ('percent', 'flat')),
discount_value      DECIMAL(10,2)      NOT NULL,
max_discount_cap    DECIMAL(10,2),      -- cap for %
discounts
min_booking_amt     DECIMAL(10,2),      -- minimum ca
rt value to apply
valid_from          DATE                NOT NULL,
valid_to            DATE                NOT NULL,
applicable_on       VARCHAR(20)        NOT NULL
CHECK (applicable_on IN ('hotel', 'transport', 'package', 'al
l')),
used_count          INT                 DEFAULT 0,
is_active           BOOLEAN             DEFAULT TRUE
);

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-- 7. BOOKING
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CREATE TABLE Booking (
booking_id          SERIAL PRIMARY KEY,
user_id            INT                 NOT NULL REFERENCES "User"(us
er_id),
booking_type       VARCHAR(20)        NOT NULL
CHECK (booking_type IN ('hotel', 'transport', 'package')),
booked_at          TIMESTAMP          DEFAULT CURRENT_TIMESTAMP,
status            VARCHAR(20)        DEFAULT 'pending'
CHECK (status IN ('pending', 'confirmed', 'cancelled', 'compl
eted')),
base_amount        DECIMAL(10,2)      NOT NULL,
discount_amount    DECIMAL(10,2)      DEFAULT 0.00,
final_amount       DECIMAL(10,2)      NOT NULL,

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payment_method VARCHAR(20) CHECK (payment_method IN ('card', 'upi', 'netbanking', 'wallet', 'cod')),
payment_ref VARCHAR(100),
promo_id INT REFERENCES Promo_Code(promo_id)
);

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-- 8. BOOKING DETAIL TABLES

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CREATE TABLE Transport_Booking (
booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON
DELETE CASCADE,
departure_id INT NOT NULL REFERENCES Schedule_Departure(departure_id),
num_seats INT NOT NULL
);

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CREATE TABLE Passenger_List (
booking_id INT NOT NULL REFERENCES Transport_Booking(booking_id)
ON DELETE CASCADE,
passenger_name VARCHAR(150) NOT NULL,
age INT NOT NULL,
gender VARCHAR(10) CHECK (gender IN ('male', 'female', 'other')),
govt_id_type VARCHAR(20) CHECK (govt_id_type IN ('aadhar', 'passport', 'voter_id', 'pan')),
govt_id_number VARCHAR(150)
);

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CREATE TABLE Hotel_Booking (
booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON
DELETE CASCADE,

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room_category_id INT NOT NULL REFERENCES Room_Category(room_c
ategory_id),
checkin_date DATE NOT NULL,
checkout_date DATE NOT NULL,
num_rooms INT NOT NULL,
num_guests INT NOT NULL
);

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CREATE TABLE Guest_List (
guest_id SERIAL PRIMARY KEY,
booking_id INT NOT NULL REFERENCES Hotel_Booking(booking_id)
ON DELETE CASCADE,
guest_name VARCHAR(150) NOT NULL,
age INT,
gender VARCHAR(10) CHECK (gender IN ('male', 'female', 'othe
r')),
govt_id_type VARCHAR(20) CHECK (govt_id_type IN ('aadhar', 'p
assport', 'voter_id', 'pan')),
govt_id_number VARCHAR(50)
);

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CREATE TABLE Package_Booking (
booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON
DELETE CASCADE,
package_departure_id INT NOT NULL REFERENCES Package_Departur
e(package_departure_id),
num_travellers INT NOT NULL
);

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CREATE TABLE Package_Traveller (
booking_id INT NOT NULL REFERENCES Package_Booking(booking_i
d)
ON DELETE CASCADE,
traveller_name VARCHAR(150) NOT NULL,
age INT NOT NULL,
gender VARCHAR(10) CHECK (gender IN ('male', 'female', 'othe

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r')),
govt_id_type VARCHAR(20) CHECK (govt_id_type IN ('aadhar', 'p
assport', 'voter_id', 'pan')),
govt_id_number VARCHAR(150)
);

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-- 9. INSURANCE
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CREATE TABLE Insurance (
insurance_id          SERIAL PRIMARY KEY,
provider_name         VARCHAR(150)    NOT NULL,
plan_name             VARCHAR(150)    NOT NULL,
coverage_summary      TEXT,
premium_amount        DECIMAL(10,2)   NOT NULL,
valid_from            DATE             NOT NULL,
valid_to              DATE             NOT NULL,
claim_status          VARCHAR(20)     DEFAULT 'none'
CHECK (claim_status IN ('none', 'filed', 'approved', 'rejecte
d'))
);

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CREATE TABLE Booking_Insurance (
booking_id           INT REFERENCES Booking(booking_id)      ON DEL
ETE CASCADE,
insurance_id         INT REFERENCES Insurance(insurance_id) ON DEL
ETE CASCADE,
PRIMARY KEY (booking_id, insurance_id)
);

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-- 10. REVIEW

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CREATE TABLE Review (
  review_id      SERIAL PRIMARY KEY,
  user_id        INT          NOT NULL REFERENCES "User"(us
er_id),
  entity_type     VARCHAR(20)   NOT NULL
CHECK (entity_type IN ('hotel', 'transport', 'package')),
  entity_id       INT          NOT NULL,
  star_rating     DECIMAL(2,1)  NOT NULL CHECK (star_rating B
ETWEEN 1.0 AND 5.0),
  title           VARCHAR(200),
  comment         TEXT,
  reviewed_at     TIMESTAMP     DEFAULT CURRENT_TIMESTAMP
);

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-- 11. CANCELLATION
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CREATE TABLE Cancellation (
  cancellation_id SERIAL PRIMARY KEY,
  booking_id      INT          NOT NULL REFERENCES Booking(b
ooking_id),
  reason          TEXT,
  cancelled_at    TIMESTAMP     DEFAULT CURRENT_TIMESTAMP,
  refund_amount   DECIMAL(10,2),
  refund_status   VARCHAR(20)   DEFAULT 'pending'
CHECK (refund_status IN ('pending', 'processed', 'failed')),
  processed_at    TIMESTAMP
);

```